ARNAUD MERCIER

Computational Geophysics
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GitLab · MODJO | GitHub · Open Seismic Hardware | Google Scholar

Doctoral researcher in Mathematical Geophysics at ETH Zurich. Research spans full-waveform inversion, experimental design, data selection, seismic imaging, scientific software and geophysical instrumentation.

Education

PhD in Computational Geophysics · Feb 2023–Present
ETH Zurich, Mathematical Geophysics | Zurich, Switzerland
Supervisor: Prof. Hansruedi Maurer, Dr. Christian Boehm
Expected completion: February 2027.

MSc in Earth Sciences and Geophysics (Graduated with Honors) · 2020–2022
Université Laval and INRS, Centre Eau Terre Environnement | Québec, Canada
Supervisors: Prof. J. Christian Dupuis, Prof. Bernard Giroux
GPA: 4.0

Bachelor of Engineering Physics (Major in Biophotonic) · 2016–2020
Université Laval | Québec, Canada
GPA: 3.9

Doctoral Research

Doctoral researcher · Feb 2023–Present
ETH Zurich, Mathematical Geophysics | Zurich, Switzerland
Research focus: full-waveform inversion, experimental design and data selection for efficient and robust geophysical inverse problems.

- Developed joint source selection and wavelet model compression for FWI, reporting approximately 50% fewer sources and 90% model compression in synthetic acoustic studies (GJI, 2025).
- Developed rank-revealing QR data selection for elastic FWI to reduce linear-system size or forward/adjoint simulation counts; evaluated crosshole and surface geometries (preprint, under review).
- Applying data selection to a real near-surface 3D survey for CO_2 storage monitoring, with an HPC workflow designed to avoid assembling the full Jacobian (ongoing).
- Developed a differentiable acoustic wave solver in JAX to investigate receiver placement through full-waveform inversion (accepted NeurIPS workshop).

- Implemented research methods in Python and shared code through MODJO; collaborating with startup Mondaic AG researchers on an upcoming open-source package.

Earlier Research Experience

Master's research in geophysical instrumentation · 2020–2022

Université Laval, Geophysical Instrumentation Group | Québec, Canada

Research focus: geophysical instrumentation and vertical seismic profiling.

- Designed seismic sensor test benches, acquisition/communication circuit boards, and microcontroller libraries; processed signals using spectral analysis, filtering, and deconvolution.
- Developed the Geophysical Open Seismic Hardware vertical profiling instrument and contributed to its hardware/firmware repository on GitHub; published the design in the Journal of Open Hardware (2023).

Scientific software and microscopy research · 2018–2019

CERVO Research Center, DCC Lab | Québec, Canada

- Developed processing and visualization software for 200 GB–2 TB microscopy datasets, with unit tests and design documentation (2018).
- Built CNN-based biological image segmentation software, a simulator based on real samples, and a Qt Designer interface (2018–2019).

Teaching Experience

Numerical Methods for Geophysics in Master of Geophysics at ETH Zurich · 2023–2026

- Designed and led practical sessions on numerical methods for geophysics.
- Designed and graded examinations.

Reflection Seismic Processing in Master of Geophysics at ETH Zurich · 2023–2026

- Supported students in reflection seismic processing, helping them break down a complex, multistep workflow into clear, manageable stages.

Geophysical methods in Geology bachelor at Laval University · 2021–2023

- Lead tutorial and Q&A session for students, grading of exercise and examinations. One on one support for programming exercise in python.

Thesis Supervision

ETH Zurich, Mathematical Geophysics · 2025–2026

Master's thesis: Experimental Characterisation of Elastic Guided Waves in Thin Plates: Toward Full-Waveform Inversion. (Main supervisor)

Bachelor's thesis: Lake-Ice Thickness Mapping Using Simulated Seismo-Acoustic Signals. (Co-supervisor)

Fieldwork Experience

3D land seismic survey (2025): Led the completion of a survey comprising 1,000 seismic nodes and 150 shots; coordinated a four-person team and liaised with contractors.

Avalanche monitoring (2024): Deployed a fiber-optic cable on an avalanche prone slope. Support in fiber layout design and anchoring system.

Reviewer Activities

Applied Computing and Geosciences, Journal of Petroleum Exploration and Production Technology, NeurIPS Workshop.

Peer Reviewed Publications

Mercier, A., Thomsen, H. R., & Moseley, B. (2026). Differentiable Physics for Sensor Placement in Full-Waveform Inversion. Sim2Science: ML with Imperfect Scientific Models, NeurIPS workshop. Open Review

Mercier, A., Boehm, C., & Maurer, H. (2025). Designing full waveform inverse problems: a combined data and model approach. Geophysical Journal International, 241(3), 1479–1494. 10.1093/gji/ggaf117

Mercier, A., Dupuis, J. C., & Giroux, B. (2023). Geophysical Open Seismic Hardware: Design of a Vertical Seismic Profiling Instrument. Journal of Open Hardware, 7(1), 3. 10.5334/joh.50

Preprints

Behnen, K., Maurer, H., Mercier, A., Hertrich, M., & Giardini, D. (2026). Combined Inversion of Active and Passive Seismic Data Based on Optimized Data Selection. Research Square. Preprint, not peer reviewed. 10.21203/rs.3.rs-10473866/v1

Mercier, A., Boehm, C., & Maurer, H. (2026). Efficient Full-Waveform Inversion via QR-Based Data Selection. EarthArXiv. Preprint, not peer reviewed. 10.31223/X5PJ4N

Conference Presentations

Mercier, A. (2026). Building Sustainable Scientific Software: Lessons from Industry–Academia. Swiss RSE Day, ETH Zurich, 31 August 2026. Accepted oral Presentation.

Mercier, A., & Maurer, H. (2024). Optimizing Full Waveform Inverse Problems: A Combined Data and Model approach. EGU General Assembly, Vienna, Austria, 14–19 April 2024. Presentation, EGU24-17729.

Mercier, A., Dupuis, J. C., & Giroux, B. (2022). Open hardware: Geophysical instrumentation for collaborations in a changing world. IMAGE, Houston, USA. Accepted oral presentation.

Mercier, A. (2022). Open seismic hardware: a step towards accessibility in mineral exploration. GeoConvention, Calgary, Canada. Accepted oral presentation.

Technical Skills And Languages

Scientific computing: Python, JAX, Optax, MATLAB, C++, C; Linux; Git CI/CD.

Numerical methods: Wave simulation, automatic differentiation, non-linear optimization, QR factorization, wavelets, experimental design.

Machine learning and data: CNN image segmentation, volumetric imaging, seismic signal processing.

Instrumentation: Sensor testing, PCB design, microcontrollers; Altium, SolidWorks, Creo, AutoCAD.

Languages: French (native), English (native) and German (proficient).

Academic Leadership And Awards

Co-founder, designer and coordinator, REDCO, Université Laval Coordinated 15 members across four optics projects; developed fiber-optic sensor prototypes and test benches (2017–2020).

Winner, “My thesis in 5 minutes”, Université Laval geology and mineral resources research center colloquium (2021).

Admission at the Autmn School on Scientific Machine Learning and Numerical Methods at CWI in Amsterdam (2025)